

Human Anatomy & Physiology

Chapter 4 — The Cardiovascular and Endocrine Systems

Braynr Open Textbook Project

4.1 The Heart and Systemic Circulation

The heart is a muscular organ roughly the size of a closed fist, located in the mediastinum between the lungs and slightly left of the body's midline. It is enclosed by a tough fibroserous sac called the pericardium and divided internally into four chambers: the right atrium, the right ventricle, the left atrium, and the left ventricle. The right side of the heart receives deoxygenated blood from the systemic circulation and propels it toward the pulmonary circuit, while the left side returns oxygenated blood to the rest of the body.

Blood flow through the heart is unidirectional, enforced by four valves. The tricuspid valve separates the right atrium and right ventricle, and the bicuspid (mitral) valve performs the analogous role on the left. Two semilunar valves — the pulmonary and aortic valves — guard the exits to the pulmonary trunk and aorta respectively. The cardiac cycle alternates between systole, the contractile phase, and diastole, the relaxation phase, producing the familiar 'lub-dub' sound when the valves close.

The conduction of the heartbeat originates in the sinoatrial (SA) node, often called the natural pacemaker, located in the wall of the right atrium. The depolarization wave then reaches the atrioventricular (AV) node, traverses the bundle of His, and propagates through the Purkinje fibers, triggering ventricular contraction. The aorta then carries oxygen-rich blood through major branches such as the brachiocephalic, left common carotid, and left subclavian arteries.

4.2 The Pancreas and Endocrine Regulation of Glucose

The pancreas is an elongated gland positioned retroperitoneally behind the stomach, with both exocrine and endocrine functions. Embedded within the pancreatic parenchyma are the islets of Langerhans, clusters of endocrine cells that secrete several hormones critical to metabolic homeostasis. Alpha cells produce glucagon, which raises blood glucose; beta cells produce insulin, which lowers it; delta cells secrete somatostatin, which suppresses both.

When circulating glucose rises after a meal, beta cells respond by releasing insulin into the portal vein, where it travels first to the liver. Insulin promotes the uptake of glucose into hepatocytes and skeletal muscle fibers, where it is converted to glycogen for storage. Conversely, during fasting the alpha cells secrete glucagon, which stimulates hepatic glycogenolysis and gluconeogenesis, releasing glucose back into the blood.

The kidney also participates in homeostasis by filtering blood within nephrons and reabsorbing virtually all filtered glucose under normal conditions. When the renal tubular maximum is exceeded, glucose appears in the urine — a clinical sign of diabetes mellitus. Persistent hyperglycemia damages microvascular beds throughout the retina, kidneys, and

peripheral nerves, leading to long-term complications.

4.3 The Brain and Neural Control of Visceral Function

Autonomic regulation of the heart, pancreas, and other visceral organs is coordinated by nuclei in the brainstem, especially the medulla oblongata, with overarching modulation from the hypothalamus. The hypothalamus integrates signals from circulating hormones, blood osmolarity, and core temperature, and projects to both the autonomic nervous system and the pituitary gland.

The pituitary gland sits in the sella turcica of the sphenoid bone, beneath the optic chiasm. Its anterior lobe synthesizes hormones such as growth hormone, ACTH, and TSH; the posterior lobe stores and releases oxytocin and antidiuretic hormone produced by hypothalamic neurons. Together, these axes make the brain–pituitary–target-gland triad the master regulator of endocrine output.

Damage to brainstem nuclei involved in cardiovascular reflexes, such as the nucleus tractus solitarius, abolishes baroreceptor responses and causes catastrophic blood pressure dysregulation. Likewise, lesions in the hypothalamic arcuate nucleus disrupt energy-balance circuits, producing hyperphagia or anorexia depending on the neuron population affected.